Name:Tharunya BS Register Number : 250701786 Experiment name: KNN

```
import numpy as np import
pandas as pd

df=pd.read_csv('Iris.csv') df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149 Data
columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----  0
     sepal.length  150 non-null    float64
1    sepal.width   150 non-null    float64
2    petal.length  150 non-null    float64
3    petal.width   150 non-null    float64 4   variety       150
nonnull    object
dtypes: float64(4), object(1) memory
usage: 6.0+ KB
df.variety.value_counts()
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
variety

Setosa        50
Versicolor    50
Virginica     50

Name: count, dtype: int64
df.head()



features=df.iloc[:,:-1].values label=df.iloc[:,4].values
```

```
from sklearn.model_selection import train_test_split from
sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2
,random_state=42)
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
KNeighborsClassifier()
```

```
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
```

```
0.9666666666666667
1.0
```

```
from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))
```

```
array([[50,  0,  0],
[ 0, 47,  3],
       [ 0,  1, 49]])
```